

HARIKA RANGISETTY

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SUMMARY

Software Engineer with 5+ years of experience building scalable microservices and high-performance backend systems across telecom, insurance, and banking domains. Specialized in Java 21, Spring Boot, event-driven architecture, and resilient API design supporting 200K+ daily transactions. Strong expertise in scalability, JVM tuning, secure authentication, and cloud deployments, consistently improving latency, availability, and system reliability.

PROFESSIONAL EXPERIENCE

Software Developer (T-Mobile) | Mphasis | USA

Feb 2025 – Present

- Engineered high-throughput, stateless microservices using Java 21 and Spring Boot, designing horizontal scaling strategies with load balancers and AWS auto-scaling groups to handle 3x normal production traffic spikes without service degradation or downtime.
- Implemented Resilience 4j circuit breaker, retry, timeout, and rate-limiting mechanisms, reducing cascading service failures across distributed systems by 60% during high-volume telecom traffic windows.
- Refined Oracle SQL queries by analyzing execution plans, eliminating full table scans, optimizing joins, and improving indexing, increasing throughput, reducing I/O contention, and optimizing infrastructure utilization.
- Migrated legacy enterprise applications to Microsoft Azure App Services, enhancing system scalability, availability, centralized monitoring, and operational resilience.
- Strengthened RESTful API security frameworks using OAuth2.0 and JWT-based authentication while implementing API rate limiting and input validation to mitigate brute-force and abuse scenarios.
- Guided junior engineers through structured code reviews, architectural discussions, and unit testing improvements, increasing test coverage to 92% using JUnit and Mockito frameworks.
- Reduced overall application memory footprint by implementing efficient object pooling strategies and stream-based processing techniques introduced in Java 21.
- Integrated secure key management solutions using AWS KMS and Azure-compatible encryption services to safeguard application secrets, credentials, and sensitive configuration data.

Software Developer (T-Mobile) | TsQs | USA

Nov 2024 – Jan 2025

- Developed scalable backend automation services in Java to orchestrate distributed test execution workflows, reducing manual regression testing efforts by 45 hours per release cycle while improving overall pipeline reliability.
- Built a predictive defect detection engine using Python and Scikit-learn, training supervised machine learning models on 20,000+ historical commit datasets to proactively reduce mean defect detection time by 14 days.
- Enhanced application runtime efficiency by integrating Redis caching layers and optimizing Docker-based deployments within AWS infrastructure, decreasing service response latency by 180ms during peak retail transaction periods.
- Constructed a centralized reporting dashboard using React.js and GraphQL APIs, providing real-time insights into test coverage metrics, defect distribution patterns, and CI pipeline health indicators.
- Reinforced backend security posture by applying OWASP secure coding standards to prevent XSS, CSRF, and SQL injection vulnerabilities across high-frequency service endpoints.
- Devised idempotent API processing patterns using unique transaction identifiers and request validation strategies to eliminate duplicate transaction handling in high-throughput financial simulations.
- Spearheaded blue-green and rolling deployment strategies across AWS-hosted environments, achieving zero-downtime production releases while minimizing rollback risks.
- Streamlined internal onboarding processes and CI/CD documentation workflows, reducing new engineer ramp-up time by 30% and improving cross-team operational consistency.

Software Developer | Liberty Mutual | USA

Feb 2024 – Oct 2024

- Re-architected legacy monolithic claim-processing modules into modular Spring Boot microservices, reducing end-to-end claim validation latency from 14 seconds to 2 seconds while improving maintainability.
- Implemented scalable event-driven pipelines using Spring Cloud and Apache Kafka, applying CAP theorem trade-off strategies to prioritize availability during peak processing of over 450,000 daily policy updates.
- Improved Hibernate ORM performance by refining caching strategies and optimizing SQL execution plans, reducing deadlocks and database contention.

- Developed reusable Angular and TypeScript component libraries with lazy loading, memoization, and optimized change detection strategies, reducing unnecessary re-render cycles across four internal insurance portals.
- Orchestrated containerized backend workloads using Docker and deployed services through AWS ECS/EKS clusters, reducing monthly production downtime by 18 hours through standardized release practices.
- Integrated real-time monitoring solutions using AWS CloudWatch and ELK stack dashboards, improving observability, log traceability, and mean-time-to-resolution during production incidents.
- Automated environment provisioning using modular Terraform Infrastructure-as-Code templates, reducing infrastructure drift issues by 80% and improving deployment repeatability.
- Hardened cloud security by configuring IAM roles, enforcing least-privilege access policies, and implementing secure key management practices across AWS-hosted services.

Assistant System Engineer | Tata Consultancy Services (TCS) | India

Jun 2019 – Oct 2022

- Built a multi-threaded backend processing engine using Java concurrency APIs and Spring Boot, reducing batch execution runtime by 140 minutes per daily processing cycle.
- Constructed a multi-threaded backend data processing engine using Java concurrency utilities and Spring Boot, reducing daily batch execution runtime by 140 minutes while improving processing reliability.
- Delivered scalable Angular-based banking modules supporting over 12,000 concurrent users, optimizing component lifecycle management, state handling, and frontend rendering performance.
- Refined Oracle SQL queries through execution plan analysis and indexing improvements, then migrated legacy applications to Microsoft Azure App Services, improving performance, scalability, availability, and operational resilience.
- Strengthened RESTful API security frameworks using OAuth2.0 and JWT-based authentication while implementing API rate limiting and input validation to mitigate brute-force and abuse scenarios.
- Guided junior engineers through structured code reviews, architectural discussions, and unit testing improvements, increasing test coverage to 92% using JUnit and Mockito frameworks.
- Reduced overall application memory footprint by implementing efficient object pooling strategies and stream-based processing techniques introduced in Java 21.
- Integrated secure key management solutions using AWS KMS and Azure-compatible encryption services to safeguard application secrets, credentials, and sensitive configuration data.

TECHNICAL SKILLS

- Programming Languages: Java (Core, Spring Boot 21), Python (Scikit-learn, Django), JavaScript, TypeScript, Go, SQL, C++, C#
- Backend & System Design: Microservices Architecture, RESTful APIs, gRPC, Event-Driven Systems, Spring Framework (Cloud, Security), Node.js, Express.js, .NET Core, OOP, Design Patterns, Scalable Distributed Systems
- Frontend & Web Development: React, Angular 14+, HTML5, CSS3, ES6+, Bootstrap, Tailwind CSS, Responsive & Mobile Web Design
- Cloud & Infrastructure: AWS (EC2, ECS, EKS, Lambda, S3, RDS, SQS, CloudFormation), Azure (SQL, Blob Storage, Data Lake, Functions), Serverless Architecture
- DevOps & CI/CD: Docker, Kubernetes (EKS/AKS), Terraform, Jenkins, GitHub Actions, GitLab CI, Maven, Bitbucket Pipelines
- Databases & Data Systems: MySQL, PostgreSQL, MongoDB, Cassandra, Redis (Caching), SQL Server, Oracle SQL, Elasticsearch, Query Optimization, Indexing
- APIs, Messaging & Integration: REST Architecture, GraphQL, Apollo, Apache Kafka, RabbitMQ, JSON, XML, API Gateway, Data Pipelines
- Testing, Monitoring & Practices: JUnit 5, PyTest, Mockito, Unit/Integration/Regression Testing, TDD, OpenTelemetry, Grafana, Prometheus, ELK Stack, Agile/Scrum, SDLC, Okta (OAuth2/OIDC)

PROJECTS

Secure API Platform for Transaction Processing

- Built a secure Python-based REST API platform processing 200,000+ daily transactions, integrating PostgreSQL and Redis caching, implementing role-based access control, structured validation, and optimized query handling across 15+ high-throughput endpoints.

Cloud-Deployed Full-Stack Monitoring Dashboard

- Developed and containerized a full-stack monitoring dashboard using Python, React, and Docker to track 25+ real-time service metrics across DEV and QA environments, supporting weekly releases with automated smoke and regression testing.

EDUCATION

Master of Science in Computer Science
Purdue University | USA

Aug 2022 – Dec 2023